

# 2003 EXT 1 July Common

## Question 1 (10 marks)

- a) i) Write an expression for  $\cos(A + B)$  (1)  
ii) Hence find the exact value of  $\cos 105^\circ$  (2)
- b) Find the coordinates of P (x, y) that divides the interval AB externally in the ratio 5:1 given A (0,2) and B (3,0) (2)
- c) Find the acute angle between the lines  $x - 2y - 6 = 0$  and  $y = -3x + 4$  to the nearest minute. (2)
- d) Prove that the line  $3x - y - 10 = 0$  is a tangent to the circle  $x^2 + y^2 = 10$  (3)

## Question 2 (8 marks) Start a new page

- a) Find i)  $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$  (1)  
ii)  $\lim_{x \rightarrow \infty} \frac{2x}{x + 2}$  (2)
- b) Solve  $2 \sin \frac{x}{2} = \cos \frac{x}{2}$  for  $0^\circ \leq x \leq 360^\circ$  (3)
- c) The minute and hour hands of a clock are 9cm and 6cm in length respectively. Find the distance between the ends of the hands when the time is 5 o'clock (to the nearest mm). (2)

## Question 3 (10 marks) start a new page

- a) Using  $\tan \frac{\theta}{2} = t$  write  $\frac{1}{2} \cot \frac{\theta}{2} - \cot \theta$  in terms of t in simplest form (2)
- b) Differentiate with respect to x,  $y = \frac{x}{3\sqrt{x}}$  (2)
- c) Prove  $\sin(\frac{\pi}{4} + A) - \sin(\frac{\pi}{4} - A) = \sqrt{2} \sin A$  (3)
- d) Solve  $\sin 2x + \cos x = 0$  for  $0^\circ \leq x \leq 360^\circ$  (3)

**Question 4 (10 marks) Start a new page**

a) Differentiate

i)  $\frac{1}{4-x^2}$  (2)

ii)  $\frac{1-x^2}{1+x^2}$  (2)

b) i) Express  $\sqrt{3} \cos x - \sin x$  in the form  $A \cos(x + \theta)$  where  $0^\circ \leq \theta \leq 90^\circ$  (2)

ii) Hence or otherwise solve  $\sqrt{3} \cos x - \sin x = 1$  where  $0^\circ \leq x \leq 360^\circ$  (2)

c) If  $y = \sqrt{r^2 - x^2}$  where  $r$  is a constant, show that  $\frac{dy}{dx} = \frac{-x}{y}$  (2)

**Question 5 (9 marks) Start a new page**

a) Find the coordinates of the points on the curve  $y = 2x^3 - 9x^2 + 27$  where the tangent is parallel to the  $x$  axis. (3)

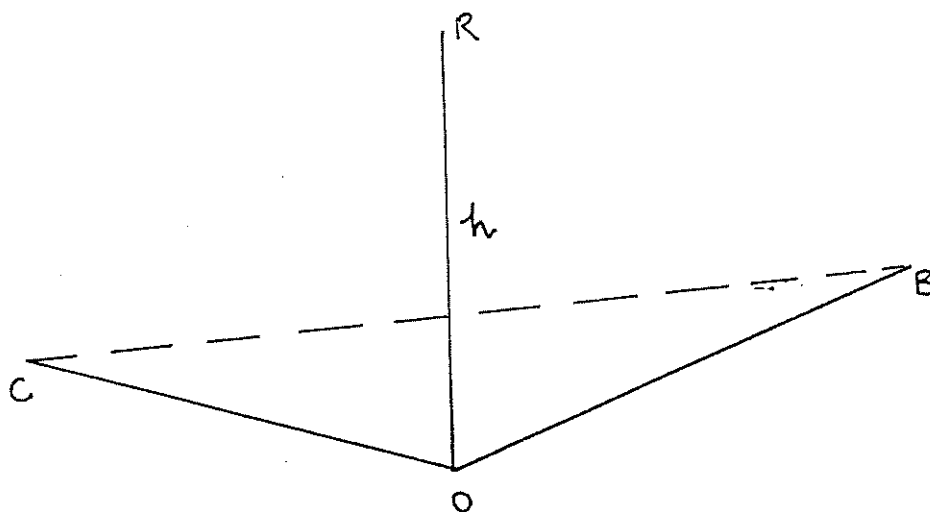
b) Given  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ , solve the equation  $\cos 3\theta + \cos \theta = 0$  for  $-180^\circ \leq \theta \leq 180^\circ$  (3)

c) i) For the function  $f(x) = x\sqrt{2-x}$  find  $f'(x)$  (2)

ii) Hence solve  $f'(x) = 0$  (1)

Question 6 (10 marks) Start a new page

- a) From a tower  $OR$ , the bearings of two points  $C$  and  $B$ , on the same level ground as the base of the tower, are  $300^\circ$  and  $015^\circ$  respectively. The angles of elevation to the top of the tower are  $11^\circ$  and  $22^\circ$  from  $C$  and  $B$  respectively.  $BC$  is  $150\text{m}$ . The tower  $OR$  has height  $h$  metres.



- i) Redraw the above diagram showing all relevant information. (1)
  - ii) Find  $\angle BOC$  (2)
  - iii) Find  $OC$  and  $OB$  in terms of  $h$  (3)
  - iv) Find the height of the tower to the nearest metre. (3)
- b) i) If  $\sin(x + \theta) = k \sin(x - \theta)$  prove  
 $(k - 1) \tan x = (k + 1) \tan \theta$  (2)
- ii) Hence solve  $\sin(x + 20^\circ) = 2 \sin(x - 20^\circ)$  if  $0^\circ \leq x \leq 360^\circ$  (2)

QUESTION 1

i)  $\cos(A+B) = \cos A \cos B - \sin A \sin B$

$\cos 105^\circ = \cos(60^\circ + 45^\circ)$   
 $= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$   
 $= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}$

$= \frac{1-\sqrt{3}}{2\sqrt{2}}$

$A(2, 2)$   $B(3, 0)$

$\left( \frac{(1 \times 0) + (-5 \times 2)}{-4}, \frac{(1 \times 2) + (-5 \times 0)}{-4} \right)$

$P\left(\frac{15}{4}, -\frac{1}{2}\right)$

$x - 2y - 6 = 0 \therefore y = \frac{x-6}{2}$

let  $m_1 = \frac{1}{2}$

$y = -3x + 4$  let  $m_2 = -3$

$\tan \theta = \left| \frac{\frac{1}{2} - (-3)}{1 + \frac{1}{2} \times (-3)} \right|$

$\tan \theta = 7$

$\therefore \theta = 81^\circ 52'$  (to nearest min)

circle  $x^2 + y^2 = 10$  has

centre  $(0, 0)$  radius  $= \sqrt{10}$

perp dist  $(0, 0)$  to  $3x - y - 10 = 0$

$r = \frac{|-10|}{\sqrt{9+1}}$

$r = \frac{10}{\sqrt{10}}$  rationalise

$r = \sqrt{10} \therefore$  since perp dist. is equal to radius  $\Rightarrow$  tangent

QUESTION 2

a) i)  $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} = 4$

ii)  $\lim_{x \rightarrow \infty} \frac{2x}{x+2} = 2$

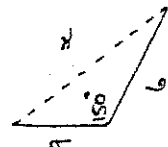
b)  $2 \sin \frac{x}{2} = \cos \frac{x}{2}$

$2 \tan \frac{x}{2} = 1$

$\tan \frac{x}{2} = \frac{1}{2}$

$\frac{x}{2} = 26^\circ 34', 206^\circ 34'$

$x = 53^\circ 8'$  in domain



$x^2 = a^2 + b^2 - 2 \times a \times b \times \cos 150^\circ$

$\therefore x = 14.5 \text{ cm (1 d.p.)}$

or  $14.5 \text{ mm}$

QUESTION 3

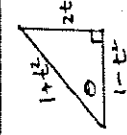
a)  $\frac{1}{2} \cot \frac{\theta}{2} - \cot \theta$

$= \frac{1}{2} \cdot \frac{1}{t} - \frac{(1-t^2)}{2t}$

$= \frac{1-1+t^2}{2t}$

$= \frac{t^2}{2t}$

$= \frac{t}{2}$



SOLUTIONS

2003 EXT 1

Question 4

a) i)  $\frac{d}{dx} (4-x^2)^{-1} = -1 \times -2x (4-x^2)^{-2}$   
 $= \frac{2x}{(4-x^2)^2}$

ii) use quotient rule

$u = 1-x^2 \quad v = 1+x^2$

$u' = -2x \quad v' = 2x$

$\frac{dy}{dx} = \frac{-2x(1+x^2) - 2x(1-x^2)}{(1+x^2)^2}$

$= \frac{-2x - 2x^3 - 2x + 2x^3}{(1+x^2)^2}$

$= \frac{-4x}{(1+x^2)^2}$

b) i)  $A = \sqrt{3+1} = 2$

$2\left(\frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x\right)$

required form

$2(\cos x \cos \theta - \sin x \sin \theta)$

$\therefore \cos \theta = \frac{\sqrt{3}}{2} \quad \sin \theta = \frac{1}{2}$

$\therefore \theta = 30^\circ$

$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x+30^\circ)$

ii)  $2 \cos(x+30^\circ) = 1$

$\cos(x+30^\circ) = \frac{1}{2}$

$x+30^\circ = 60^\circ, 300^\circ$

$x = 30^\circ, 270^\circ$

c)  $y = (r^2 - x^2)^{-1/2}$

$\therefore \frac{dy}{dx} = \frac{1}{2} \times -2x (r^2 - x^2)^{-3/2}$

$= \frac{-x}{\sqrt{r^2 - x^2}}$

since  $-\frac{x}{y} = \frac{-x}{\sqrt{r^2 - x^2}}$

$\therefore \frac{dy}{dx} = -\frac{x}{y}$

b)  $y = \frac{1}{3} x \cdot x^{1/2}$

$\therefore y = \frac{1}{3} x^{3/2}$

$\frac{dy}{dx} = \frac{1}{3} \times \frac{3}{2} x^{1/2}$

$= \frac{1}{2} \sqrt{x}$

c) Prove  $\sin\left(\frac{\pi}{4} + A\right) - \sin\left(\frac{\pi}{4} - A\right) = \sqrt{2} \sin A$

LHS  $= \sin \frac{\pi}{4} \cos A + \cos \frac{\pi}{4} \sin A - \left(\sin \frac{\pi}{4} \cos A - \cos \frac{\pi}{4} \sin A\right)$

$= \frac{1}{\sqrt{2}} \cos A + \frac{1}{\sqrt{2}} \sin A - \frac{1}{\sqrt{2}} \cos A + \frac{1}{\sqrt{2}} \sin A$

$= \frac{2}{\sqrt{2}} \sin A$

$= \frac{2 \times \sqrt{2}}{\sqrt{2}} \sin A$

$= \frac{2\sqrt{2}}{2} \sin A$

$= \sqrt{2} \sin A$

$= \text{RHS}$

d)  $\sin 2x + \cos x = 0$

$2 \sin x \cos x + \cos x = 0$

$\cos x (2 \sin x + 1) = 0$

$\cos x = 0 \quad \sin x = -\frac{1}{2}$

$\therefore x = 90^\circ, 270^\circ \quad \text{acute } x = 30^\circ \quad \therefore x = 210^\circ, 330^\circ$

### Question 5

a)  $y = 2x^3 - 9x^2 + 27$

$$\frac{dy}{dx} = 6x^2 - 18x$$

$$\frac{dy}{dx} = 0 \quad \therefore 6x^2 - 18x = 0$$

$$6x(x - 3) = 0$$

$$\therefore x = 0 \quad x = 3$$

$\therefore (0, 27) \quad (3, 0)$

b)  $\cos 3\theta + \cos \theta = 0$

$$4\cos^3 \theta - 3\cos \theta + \cos \theta = 0$$

$$4\cos^3 \theta - 2\cos \theta = 0$$

$$2\cos \theta (2\cos^2 \theta - 1) = 0$$

$$\therefore \cos \theta = 0 \quad \cos \theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = \pm 90^\circ, \pm 45^\circ, \pm 135^\circ$$

c) i)  $f(x) = x \sqrt{2-x}$

$$u = x \quad v = \sqrt{2-x} = (2-x)^{1/2}$$

$$u' = 1 \quad v' = -\frac{1}{2}(2-x)^{-1/2}$$

$$= \frac{-1}{2\sqrt{2-x}}$$

$$f'(x) = \sqrt{2-x} - \frac{x}{2\sqrt{2-x}}$$

ii)  $\sqrt{2-x} - \frac{x}{2\sqrt{2-x}} = 0$

$$2(2-x) - x = 0$$

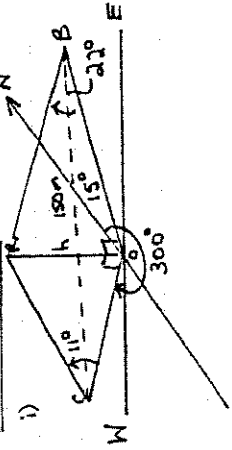
$$2\sqrt{2-x}$$

$$4 - 2x - x = 0$$

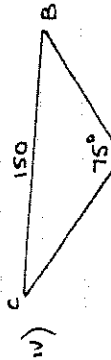
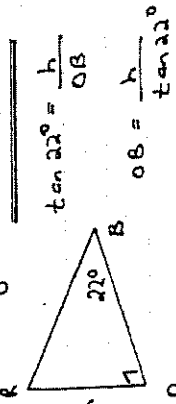
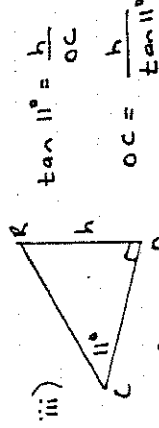
$$4 - 3x = 0$$

$$\therefore x = \frac{4}{3}$$

### Question 6



ii)  $\angle BOC = 75^\circ$



use cosine rule

$$150^2 = h^2 + h^2 - \frac{h^2}{\tan^2 22^\circ \tan^2 11^\circ}$$

$$\left(2 \frac{h}{\tan 22^\circ \tan 11^\circ} \cos 75^\circ\right)$$

$$150^2 = h^2 \left[ \frac{1}{\tan^2 22^\circ \tan^2 11^\circ} + \frac{1}{\tan^2 22^\circ \tan^2 11^\circ} - \frac{2 \cos 75^\circ}{\tan^2 22^\circ \tan^2 11^\circ} \right]$$

$$h = 29 \text{ m (nearest metre)}$$

b) i)  $\sin(x+\theta) = k \sin(x-\theta)$

$$\sin x \cos \theta + \cos x \sin \theta = k [\sin x \cos \theta - \cos x \sin \theta]$$

$$\sin x \cos \theta + \cos x \sin \theta = k \sin x \cos \theta - k \cos x \sin \theta$$

$$\div \text{ by } \sin x \cos \theta$$

$$1 + \frac{\cos x \sin \theta}{\sin x \cos \theta} = k - k \frac{\cos x \sin \theta}{\sin x \cos \theta}$$

$$1 + \frac{\tan \theta}{\tan x} = k - k \frac{\tan \theta}{\tan x}$$

$$\tan x + \tan \theta = k \tan x - k \tan \theta$$

$$\tan \theta + k \tan \theta = k \tan x - \tan x$$

$$(k+1) \tan \theta = (k-1) \tan x$$

$$\therefore k = 2 \quad \theta = 20^\circ$$

ii)  $\sin(x+20^\circ) = 2 \sin(x-20^\circ)$

$$3 \tan 20^\circ = \tan x$$

$$1.091 \dots = \tan x$$

$$\therefore x = 47^\circ 31', 227^\circ 31'$$

|   |   |   |
|---|---|---|
| S | A | V |
| V | T | C |